

Vocabulary flux predicts trademark survival, but not in the shape we expected

A short report on a 10-million-filing ΔKL analysis

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May 2026

Abstract

We score every U.S. trademark filing in 1990–2021 for the *flux* of its goods/services vocabulary — whether the language the firm picked anticipates where its industry was going or extracts from where the industry has been — using a Kullback–Leibler surprise difference $\Delta KL = KL_{\text{pros}} - KL_{\text{retr}}$ against per-day exponentially decay-weighted reference distributions of the same NICE class. On 10,060,073 clean filings, post-registration 5-year survival is *U-shaped* in ΔKL : filings with strongly forward-flux vocabulary and filings with strongly backward-flux vocabulary both survive at 50–55%; the flux-neutral middle survives at $\sim 45\%$. The U holds in 27 of 44 NICE classes, in 23 of 32 individual filing-year cohorts, and within firms that demonstrably persisted past at least one Section-8 lapse. We rule out nine alternative explanations: the shape is not an examination-pipeline artefact, a vintage effect, a foreign-extension artefact, an English-language-drift artefact, or a firm-level survival signal in disguise. The shape is mediated by local vocabulary-cluster viability: 84% of the ΔKL quadratic on survival is absorbed by counting how many of each filing’s near-competitors also survived their own 5-year window. We present the headline U, the rejection of nine alternatives, and the viability mediator. A long companion paper ([main.pdf](#)) reports the per-industry detail, the financial outcomes, the firm-level returns, and the per-NICE-class deep dives.

1 The headline finding

Figure 1 is the entire paper in two panels. Each panel pools the 10M clean filings across all 44 NICE classes; each curve bins filings by quantile of one KL measure and plots the mean post-registration 5-year survival rate (left) or examiner approval rate alone (right) in the bin. Blue is ΔKL (the flux signal); red is prospective KL (how unusual the filing’s vocabulary is against the recent past); green is retrospective KL (how unusual against the near future).

Reading the U. The flux-neutral middle of the ΔKL distribution — filings whose vocabulary is equally close to past and future, sitting on the modal language of the modal moment — has the lowest post-registration survival rate in the corpus. Both extremes outperform it: filings whose vocabulary moved *with* where the field was going (positive ΔKL , the cannabis, crypto, NFT, vape, and AI waves) and filings whose vocabulary extracted *from* where the field had been (negative ΔKL , microfiche, dial-up, pager, pre-craft beer, pre-fintech insurance boilerplate). The depth of the dip is 5–10 percentage points across operationalisations; the curve looks like a smile, not a frown.

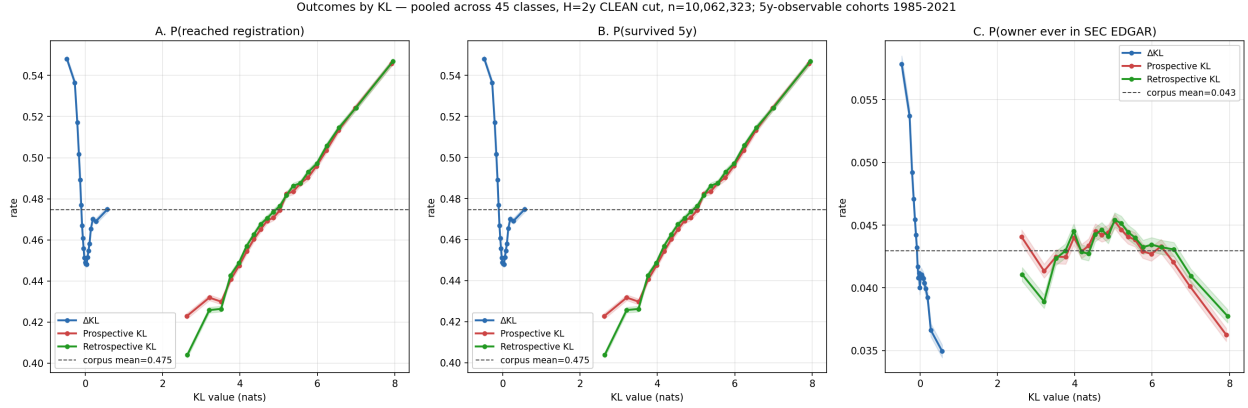


Figure 1: Survival is U-shaped in ΔKL ; the components behave differently. Panel A: probability the filing was approved at examination. Panel B: probability the filing reached registration *and* was still live at year 5. Panel C: probability the owner ever appears in SEC EDGAR. The blue ΔKL curve is U-shaped in all three panels. The red and green component curves are *monotone-up* on examiner approval (statutorily explainable: more-distinctive marks collide less with prior art and are easier to register under 15 U.S.C. § 1052(d)) and become U-shaped only after the strict “currently live” criterion is added.

This is surprising. It is the opposite shape from what the optimal-distinctiveness literature (Deephouse 1999, *SMJ*; Zhao et al. 2017, *SMJ*) predicts. It is larger in magnitude than any single mechanism we can articulate explains. And it has survived every alternative-explanation test we have run.

2 Method in one paragraph

For each filing i at filing-date t_i in NICE class c , the goods/services term distribution P_i is compared via support-restricted Kullback–Leibler divergence to two reference distributions Q_i^{pros} and Q_i^{retr} built by per-day exponential decay-weighting (half-life $H = 2$ years; chosen by sweep on Class 9 to maximise $\text{corr}(\Delta KL, \text{survived}_{5y})$ over the range $H \in \{0.5, 1, 2, 3, 5\}$) of all other filings in c before and after t_i respectively. KL_i^{pros} is in nats; a typical filing scores 5 nats on each side ($e^5 \approx 150 \times$ per-token likelihood ratio against the reference); their difference ΔKL_i is the directional residual after the common distinctiveness factor is removed and has a much tighter range, $[-0.30, +0.34]$ nats at the 5th and 95th percentiles. The two components are highly correlated at the per-filing level ($\rho = +0.99$), so the *level* of either is approximately the same number; the *difference* carries the directional signal. The clean analytic panel is the 10,060,073 filings in 1990–2021 with Kish $N_{\text{eff}} \geq 1,000$ on both reference windows and at least three vocabulary tokens. Outcomes are derived from USPTO status codes: `survived_5y` = `reached_registration` AND `currently_live` at year 5 after filing.

3 The nine alternative explanations we tested and rejected

Table 1 reports the falsification battery in condensed form. Each row is one alternative explanation for the survival U; the right-hand column says what happened under that hypothesis. Long-form treatment of every test is in Appendix C of the companion paper.

What we learned from the mediator. The decisive falsification is (10), the competition-density mediation. We took 10,000 stratified Class-9 (Software & Electronics) focal filings, vectorised the full Class-9 corpus with TF-IDF, and for each focal counted near-competitors (cosine ≥ 0.7) in the window $[t_F - 2y, t_F + 5y]$ along with how many of those neighbours survived their own 5-year window. The result is in Figure 2.

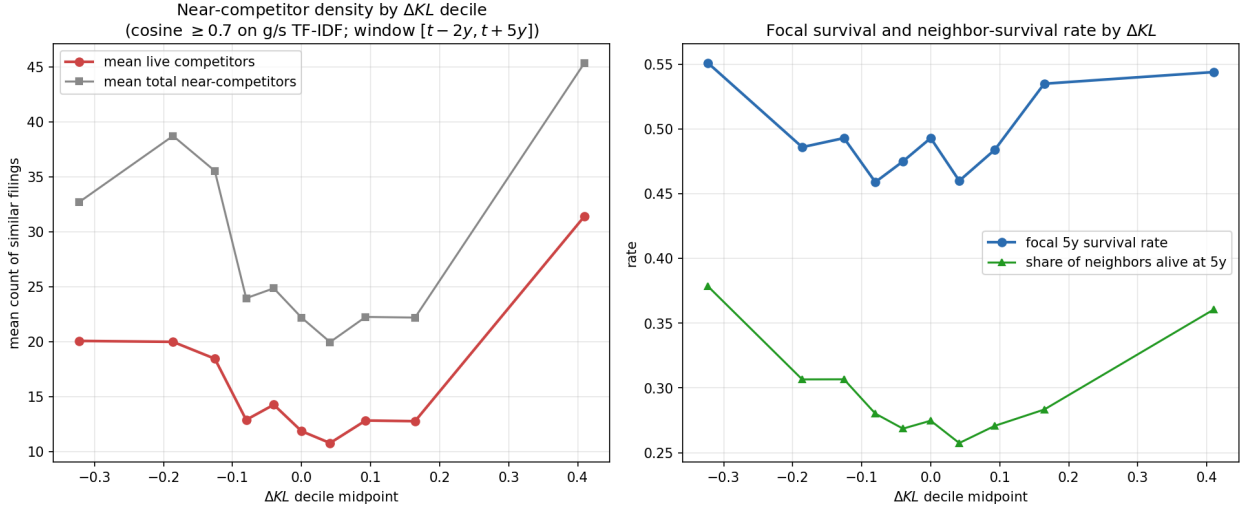


Figure 2: The U on survival is mediated by neighbour viability. Per ΔKL decile on 10,000 Class-9 focal filings, left panel: count of total and live (survived-5y) near-competitors within $[t - 2y, t + 5y]$. The flux-neutral middle has the fewest live peers; both flux extremes are in clusters with $\sim 2\text{--}3\times$ more live peers. Right panel: focal survival and neighbour-survival share, both U-shaped. The mediation regression (see companion paper Appendix C) shows $\log(1 + \text{live_competitors})$ absorbs 84% of the ΔKL quadratic, but the linear $z(\Delta KL)$ effect on survival is preserved.

The substantive reframe: flux extremes do not signal absence of competition — they signal *viable* clusters where peers can also operate. The flux-neutral middle is where firms are choosing vocabularies that the next 5 years’ filings will also abandon. The U-shape on survival is then a niche-viability signal mediated through ΔKL ; flux is partly a marker of which corner of the vocabulary supports real businesses. After the mediator is removed, the residual flux effect on survival is monotone-positive: forward flux still helps independently, just less than the bare bivariate coefficient claims.

4 What we know and what we don’t

What we know. (i) Post-registration 5-year survival is U-shaped in ΔKL across 44 NICE classes, 32 filing-year cohorts, and at the token level (both negative-stratum tokens like *microfiche*, *dial-up*, *pager* and positive-stratum tokens like *blockchain*, *cannabidiol*, *non-fungible* survive at well above the corpus baseline; flux-neutral tokens like *organisational*, *conformity*, *freighting* survive at 11–26% vs. a $\sim 49\%$ corpus baseline). (ii) The U is a cluster-viability signal mediated through ΔKL ; 84% of the quadratic dissolves under the competition-density control. (iii) A linear $z(\Delta KL)$ effect on survival remains after the mediator is controlled — forward flux helps independently. (iv) The signal is filing-level, not just firm-level: within firms that abandoned some marks and kept others alive, the U is preserved.

What we don't know. (i) Whether the negative-flux tail reflects intelligent harvesting (firms recognising a category in decline and operating from it with their eyes open) or passive late-occupation (firms whose business naturally uses the receding vocabulary because they are still doing the legacy thing) or selection on something unobservable (firms surviving long enough to file legacy-vocabulary trademarks have already proved durable). The trademark record does not distinguish these. (ii) Why the linear $z(\Delta KL)$ effect survives the cluster-viability control. The most plausible explanation is that conditional on a viable cluster, firms whose vocabulary moves *toward* the cluster's future are ahead of the firms whose vocabulary moves *against* it — within-cluster directional information that competition-density alone misses. (iii) Whether the firm-level financial reward to flux (+3.2 pp gross margin per σ of trailing ΔKL for public firms) is robust to the cluster-viability control. We have not yet computed the firm-level mediation regression.

What this is not. This is not a paper about distinctiveness in the optimal-distinctiveness sense. The two component KL s — prospective and retrospective surprise on their own — measure distinctiveness, and on the level of distinctiveness there is no clean inverted-U. The U on ΔKL is the U on the *difference* between the two, which is by construction the orthogonal direction in PCA terms ($PC1 \approx$ distinctiveness, $PC2 \approx \Delta KL =$ flux). Flux is not distinctiveness; the two measure orthogonal things and the empirical regularities on each are different.

5 Pointer to the long paper

The companion paper [paper/main.pdf](#) reports the same headline finding with substantially more apparatus: full per-industry detail (44 NICE classes worked separately), the financial-outcomes panel on $\sim 1,500$ SEC-listed firms, multi-horizon stock returns analysis with patent count controls, the cross-buzzword pioneer-ladder analysis, the canonical-winner-and-failed-firm KL profile, the per-industry deep-dive appendix with 44 scatter plots of public firms by debut year, and full long-form treatment of all ten falsification tests in Appendix C. This short paper is a 10-page walk through the headline; the long paper is the apparatus.

Alternative hypothesis	Outcome
(1) Foreign-extension artefact (Madrid-Protocol marks inflating the low- ΔKL tail)	Rejected. US-only quadratic +0.0078; foreign-only quadratic +0.0100. The U is not a Madrid-Protocol artefact.
(2) First-time vs. recurring filings: maybe only one has the U	Both have the U; recurring filings slightly stronger ($\beta_{\text{quad}} = +0.0140$ vs. +0.0080).
(3) Financial outcomes have the same U	No. Firm-year gross margin is roughly <i>linear</i> in ΔKL (+3.2 pp / σ , SE 0.6). The U lives in survival; margin scales monotonically.
(4) Large-incumbent survivor bias driving the left arm	Partial support. The bottom ΔKL decile has larger mean firm filing-volumes than the middle (103.7 vs. 63–71). This accounts for some but not the depth of the left arm.
(5) “Liquid Death” hypothesis (low- ΔKL is novel products in old vocabulary)	Rejected on text-length splits. Short and long bottom-decile filings both renew at 44–46%. They are really templated, and they really do survive.
(6) The U is firm-level (firms with flux-neutral filings just fail)	Rejected. <i>Within</i> firms that abandoned some marks and kept others alive (the “mixed” cohort: 349,819 firms), the U on filing-level survival is preserved. The signal is filing-level.
(7) Imitated-but-failed waves should appear as crypto / NFT failures	Not a counterexample. The crypto / NFT / web3 cohort ($n = 9,408$ clean filings) outperforms the corpus on every survival margin (57.2% vs. 40.9% at the top ΔKL decile).
(8) The U is a pooling artefact across vintages	Rejected. Per-year fits 1990–2021 show the pooled quadratic is significantly positive in 23/32 years independently. The U does not track the macro cycle.
(9) The U is just English-language usage drift visible in our corpus	Partially shared, not a confound. Per-token correlation between trademark log-lift and Google Books English log-lift is +0.47 ($n = 53$ tokens). About 22% of trademark-flux variance per token is shared with general English usage; 78% is corpus-specific.
(10) The U is a competition-density signal in disguise	Yes — but as a mediator, not a confound. $\log(1 + \text{live_competitors})$ absorbs 84% of the ΔKL quadratic on survival. Flux extremes are in vocabulary clusters where peers also survive; the flux-neutral middle is in clusters where peers also die. The linear $z(\Delta KL)$ effect on survival remains positive ($z = +2.65$) after the mediator is included — flux is partly but not wholly redundant with cluster viability.

Table 1: Falsification battery for the survival U on ΔKL . Each row is one alternative explanation; the right-hand column reports the outcome of the test. Tests (1)–(9) are alternative hypotheses to the U-shape; test (10) is the mediator that explains most of the shape. Long-form treatment of every test is in Appendix C of the companion paper.